



Ahsanullah University of Science and Technology Bangladesh

COURSE OUTLINE

1. Title: **Soft Computing**
2. Code: **CSE4237**
3. Credit hours: **3**
4. Level: **Level 4, Term 2**
5. Faculty: **Engineering**
6. Department: **Computer Science and Engineering (CSE)**
7. Programme: **Bachelor of Science in Computer Science and Engineering (B.Sc. in CSE)**
8. Synopsis from the Approved Curriculum:

Aims and constituents of soft computing. Fuzzy sets and logic: Concepts and properties of fuzzy sets; Mathematical & logical implications of fuzzy sets; Fuzzy relations; Applications of fuzzy sets in information processing, decision making and control systems. Artificial neural networks: Underlying ideas and concepts of artificial neural networks; Feed-Forward, Recurrent and other types of artificial neural networks; Rules and methodologies of training artificial neural networks; Error backpropagation, recurrent backpropagation and other learning algorithms for neural networks. Probabilistic reasoning: Bayesian inference models and Bayesian networks; Dempster – Shafer theory; Probabilistic decision support systems. Genetic algorithms: Underlying principles and fundamental operators of genetic algorithms; Searching based on genetic algorithms; Genetic algorithm based optimization, learning and control. Introduction to various neuro-fuzzy-probabilistic-genetic combined approaches to computing applications.

9. Type of course (core/elective): **Elective**
10. Prerequisite(s) (if any): **CSE3101 (Mathematical Analysis for Computer Science)**
11. Name of the instructor(s) with contact details and office hours:

Nibir Chandra Mandal

Room: 7A01/O
 Phone: N/A
 E-mail: nibir.cse@aust.edu
 Office hour: TUE 11:20 AM – 01:00 PM, THU 10:30 AM – 12:10 PM

12. Semester Offered: **Fall, 2021**

13. Mapping of Course Outcomes with Bloom's Taxonomy and Programme Outcomes

After successful completion of the course, the students will be expected to:

Sl. No.	COs	POs	Bloom's Taxonomy		
			C	A	P
1	Comprehend the fundamental concepts used in Soft computing. The concepts of Fuzzy logic (FL), major technology trends driving Deep Learning and optimization techniques using Genetic Algorithm (GA).	1	2		
2	Apply proven techniques from soft computing and probabilistic model to solve computational, mathematical and probabilistic problems, ideas and concepts of artificial neural networks; Feed-Forward, Recurrent and other types of artificial neural networks.	2	3		
3	Analyze complex problems from probability models and soft computing for prospective and alternative solutions, rules and methodologies of training artificial neural networks; Error backpropagation, recurrent backpropagation and other learning algorithms for neural networks.	4	4		
4	Design novel and hybrid techniques to solve complex problems from soft computation; sequence models to natural language processing and computer vision problems.	3	6		

14. Mapping of COs with Knowledge Profiles, Complex Engineering Problem Solving and Complex Engineering Activities

Course Outcome	Knowledge Profile	Complex Problem Solving	Complex Engineering Activities
CO1	K3		
CO2	K3		
CO3	K4		
CO4	K6		
CO5	K5		

15. Percentages of Assessment Methods

Method	Percentage
Class Performance	10
Quizzes	30
Final Examination	60

16. Week wise distribution of contents and assessment methods

Week	Topics	Assessment Method(s)
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1	<i>MAP: The need for probabilistic predictions, Introduction to deep learning, Why is deep learning taking off?</i>	
2	<i>Formal Foundation for Making Probabilistic Models: HMMs, Viterbi Algorithm, Neural Networks Basics, Derivatives, Computation graph, Derivatives with a Computation Graph.</i>	
3	<i>Genetic algorithms: Underlying principles and fundamental operators of genetic algorithms, Logistic Regression Gradient Descent, Vectorization, Vectorizing Logistic Regression, Shallow neural networks, Computing a Neural Network's Output, Vectorizing across multiple examples.</i>	Quiz-1
4		
5		
	<i>Searching based on genetic algorithms, Activation functions, Derivatives of activation functions, Gradient descent for Neural Networks.</i>	

6	<i>Genetic algorithm based optimization: Travelling Salesman Problem, Ant Colony Optimization, Deep Neural Networks, Deep L-layer neural network, Building blocks of deep neural networks.</i>	
7		
8	<i>Foundations of Fuzzy Approaches: Fuzzy sets, Basic Terms and Operations, Properties of Fuzzy Sets, Extension Principle, Alpha-Cuts, Forward and Backward Propagation, Parameters vs Hyperparameters, regularization, overfitting, Dropout Regularization.</i>	Quiz 2
9		
10	<i>Fuzzy Relations: Properties of Fuzzy Relations, Basic Operations with Fuzzy Relations, Vanishing / Exploding gradients, Numerical approximation of gradients, Gradient checking implementation notes.</i>	

11	<i>Fuzzy Numbers: Representing Fuzzy Numbers, Fuzzy Numbers: Addition, Subtraction and Multiplication, Natural Language Processing & Word Embeddings, Learning Word Embeddings: Word2vec , Applications using Word Embeddings.</i>	
12	<i>Integrated Neural-Fuzzy Technology: Generalized Fuzzy Neuron and Networks, AND and OR Fuzzy Neurons, Recurrent Neural Networks, Sequence models & Attention mechanism.</i>	Quiz 3
13		
14	Review classes.	

17. References

17.1. Required (if any)

- 1. Neuro-Fuzzy and Soft Computing: A Computational Approach to Learning and Machine Intelligence (1st Edition)***

Authored by: Jyh-Shing Roger Jang, Chuen-Tsai Sun and Eiji Mizutani

Publisher: Pearson

- 2. Fuzzy and Neural Approaches in Engineering.***

Authored by: Lefteri H. Tsoukalas.

Publisher: Wiley-Interscience; 1 edition (February 5, 1997).

- 3. Dive into Deep Learning***

Authored by: Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola

- 4. Deep Learning***

Authored by: Ian Goodfellow and Yoshua Bengio and Aaron Courville

Publisher: MIT Press

17.2. Recommended (if any)

- 1. Course website - <https://classroom.google.com/c/NTE2MDIzODUzNTVa>**
- 2. Essentials of Metaheuristics (Online Version 2.2), Sean Luke, Department of Computer Science, George Mason University -**

<https://cs.gmu.edu/~sean/book/metaheuristics/Essentials.pdf>

- 3. Course**

- 4. Course.**

Prepared by:	Checked by:	Approved by:
Signature: _____	Signature: _____	Signature: _____
Name: Nibir Chandra Mandal Department: CSE Date:	Name: Dr. Mohammad Shafiul Alam OBE Program Coordinator, CSE Date:	Name: Dr. Mohammad Shafiul Alam HOD, CSE Date:

Annex-1: PEO of CSE

PEO1 - Professionalism

Graduates will demonstrate sound professionalism in computer science and engineering or related fields.

PEO2 – Continuous Personal Development

Graduates will engage in life-long learning in multi-disciplinary fields for industrial and academic careers.

PEO3 – Sustainable Development

Graduates will promote sustainable development at local and international levels.

Annex-2: Mapping of PEO-PO

	PEO1	PEO2	PEO3
PO1 - Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.	√		
PO2 - Problem analysis: Identify, formulate, research and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.	√		
PO3 - Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.	√		
PO4 – Investigation: Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.	√		

PO5 - Modern tool usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.	√		
PO6 - The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.	√		√
PO7 - Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.	√		√
PO8 – Ethics: Apply ethical principles and commit to professional ethics, responsibilities and the norms of engineering practice.	√		
PO9 - Individual work and teamwork: Function effectively as an individual and as a member or leader of diverse teams as well as in multidisciplinary settings.	√	√	
PO10 – Communication: Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear	√		

instructions.			
PO11 - Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work as a member or a leader of a team to manage projects in multidisciplinary environments.	√		
PO12 - Life-long learning: Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.		√	

Annex-3: Blooms Taxonomy *

Level	Cognitive Domain – Revised Version	Affective Domain	Psychomotor Domain
1	Remember (1)	Receiving Phenomena (1)	Perception (1)
2	Comprehend (2)	Responding to Phenomena (2)	Set (2)
3	Apply (3)	Valuing (3)	Guided Response (3)
4	Analyse (4)	Organizing Values (4)	Mechanism (4)
5	Evaluate (5)	Internalising Values (5)	Complex Overt Response (5)
6	Create (6)		Adaption (6)
			Origination (7)

* Based on “REVISED BLOOM’S TAXONOMY INDICATOR v3.31” , available at <http://adept.mmu.edu.my/wp-content/uploads/2018/09/Blooms-Taxonomy-Indicator-v3.31.xls>

Annex-4: Knowledge Profile

BAETE MANUAL 2019, 2 nd ed. (TABLE 4.1) - KNOWLEDGE PROFILE		
K1	Natural sciences	A systematic, theory-based understanding of the natural sciences applicable to the discipline.
K2	Mathematics	Conceptually-based mathematics, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modelling applicable to the discipline.
K3	Engineering fundamentals	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
K4	Specialist Knowledge	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
K5	Engineering Design	Knowledge that supports engineering design in a practice area.
K6	Engineering Practice	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
K7	Comprehension	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline; ethics and the professional responsibility of an engineer to public safety; the impacts of engineering activity: economic, social, cultural, environmental and sustainability.
K8	Research literature	Engagement with selected knowledge in the research literature of the discipline.